

JSS1 Mathematics Diagnostic Bank (60 Questions)

Scope: Primary 5/6 Foundation & Full JSS1 Curriculum (Cambridge & Nigerian)

Focus: Terminal Diagnostic & JSS2 Readiness

Cluster 1: Number Sense & Operations (Questions 1 - 20)

No.	Question & Options	Ans	Source & Diagnostic Purpose
1	Round 45,678 to the nearest hundred. A) 45,600 B) 45,700 C) 46,000 D) 45,000	B	[P5/6] Tests foundational understanding of place value and rounding.
2	What is the value of 8 in 4.582? A) 8 tenths B) 8 hundredths C) 8 thousandths D) 8 units	B	[P6/JSS1] Decimal place value. High error in students struggling with fractions.
3	Simplify: $\$(-15) + (+7) - (-3)\$$ A) -11 B) -5 C) 19 D) -25	B	[JSS1] Basic Integers. Tests sign manipulation readiness for JSS2 Algebra.
4	Express 0.45 as a fraction in its simplest form. A) 45/100 B) 9/20 C) 4/5 D) 9/25	B	[P6/JSS1] Converting decimals to fractions.
5	Find the LCM of 12, 18, and 24. A) 6 B) 36 C) 72 D) 144	C	[P6] Foundational skill for solving complex algebraic fractions later.
6	Calculate: 25% of 480. A) 120 B) 96 C) 240 D) 12	A	[P5/6] Percentage application.
7	A car travels 180km in 3 hours. What is its speed in km/h? A) 60 B) 540 C) 90 D) 183	A	[P6/JSS1] Rate and Ratio foundation.
8	Write 72 as a product of its prime factors in index form. A) $2 \times 2 \times 2 \times 3 \times 3$ B) $2^3 \times 3^2$ C) 8×9 D) $2^2 \times 3^3$	B	[JSS1] Prime factorization and Indices introduction.
9	Solve: $\$3/4 - 1/8 + 1/2\$$ A) 3/8 B) 1 1/8 C) 7/8 D) 5/8	B	[P6] Operations with fractions (unequal denominators).
10	Complete the sequence: 1, 4, 9, 16, ... A) 20 B) 24 C) 25 D) 32	C	[Cambridge JSS1] Pattern recognition and Square numbers.
11	Highest Common Factor (HCF) of 45 and 75 is: A) 5 B) 15 C) 3 D) 225	B	[P6] Factorization skills.
12	Which of these is a prime number? A) 1 B) 9 C) 21 D) 37	D	[P5] Identification of prime numbers.
13	Evaluate: $\$ \sqrt{144} + (3^3 \div 9) \$$ A) 15 B) 18 C) 21 D) 13	A	[JSS1] Combined operations (BODMAS/PEMDAS) with squares and indices.
14	Change 4/5 to a percentage. A) 40% B) 80% C) 45% D) 0.8%	B	[P6] Equivalent conversion skills.
15	Standard form of 0.00045 is: A) 4.5×10^{-4} B) 4.5×10^4 C) 45×10^{-5} D) 0.45×10^{-3}	A	[Cambridge JSS1] Introduction to scientific notation.
16	Arrange in ascending order: 0.7, 3/4, 65%. A) 65%, 0.7, 3/4 B) 0.7, 65%, 3/4 C) 3/4, 0.7, 65% D) 65%, 3/4, 0.7	A	[JSS1] Comparing different number formats.
17	Sharing 500 Naira in ratio 2:3. Smaller share is: A) 200 B) 300 C) 100 D) 250	A	[JSS1] Practical Ratio application.
18	Calculate the square root of 0.81. A) 0.09 B) 9 C) 0.9 D) 8.1	C	[JSS1] Decimal squares. Common misconception check.

No.	Question & Options	Ans	Source & Diagnostic Purpose
19	A watch was bought for N2000 and sold for N2500. Calculate % profit. A) 20% B) 25% C) 50% D) 5%	B	[P6/JSS1] Business Math foundation.
20	Simplify: $2^5 \div 2^2$ A) 2^7 B) 2^3 C) 4 D) 8	B	[JSS1/Cambridge] Laws of Indices (Division).

Cluster 2: Algebraic Processes (Questions 21 - 35)

No.	Question & Options	Ans	Source & Diagnostic Purpose
21	Simplify: $4a + 3b - a + 2b$. A) $3a + 5b$ B) $5a + 5b$ C) $4a + 5b$ D) $7ab$	A	[JSS1] Collecting like terms. Essential Algebra foundation.
22	Solve for x: $3x - 5 = 10$ A) 5 B) 15 C) 3 D) 2	A	[JSS1] Simple linear equations (one-step).
23	If $a=2$ and $b=3$, evaluate: $2a^2 + b$. A) 11 B) 7 C) 19 D) 15	A	[JSS1] Substitution in algebraic expressions.
24	Translate to algebra: "5 less than twice a number n". A) $5 - 2n$ B) $2n - 5$ C) $2(n-5)$ D) $5 < 2n$	B	[JSS1/Cambridge] Mathematical modeling skills.
25	Simplify: $3(x + 4) - 2x$. A) $x + 4$ B) $x + 12$ C) $5x + 12$ D) $x + 7$	B	[JSS1] Removal of brackets (Distributive law).
26	Solve: $x/4 = 8$. A) 2 B) 12 C) 32 D) 4	C	[P6/JSS1 Bridge] Inverse operations (Multiplication/Division).
27	Factorize: $4x + 12y$. A) $4(x + 3y)$ B) $16xy$ C) $4xy$ D) $2(2x + 6y)$	A	[JSS1] Common factor extraction.
28	Solve for y: $2(y - 3) = 14$. A) 10 B) 7 C) 4 D) 17	A	[JSS1] Multi-step linear equations.
29	Expression for perimeter of rectangle with length L, width W. A) LW B) $L+W$ C) $2(L+W)$ D) L^2W	C	[P6/JSS1] Formulas in algebra.
30	If 3 is added to a number, the result is 11. What is the number? A) 14 B) 8 C) 33 D) 1.5	B	[P5/6 Bridge] Word problem translation.
31	Which is an equation? A) $3x + 4$ B) $5y < 2$ C) $2x = 10$ D) $4x - 7z$	C	[JSS1] Identifying algebraic structures.
32	Simplify: $x * x * x$. A) $3x$ B) x^3 C) $3x^3$ D) $x + 3$	B	[JSS1] Power notation in algebra.
33	Evaluate $4x-y$ when $x=0$, $y=5$. A) 5 B) -5 C) 4 D) 0	B	[JSS1] Working with Zero and Negatives in substitution.
34	Coefficient of x in the expression $5 - 3x$ is: A) 5 B) 3 C) -3 D) x	C	[JSS1 Terminology] Identifies parts of an expression.
35	Solve: $0.5x = 10$ A) 5 B) 20 C) 2 D) 10.5	B	[JSS1/Cambridge] Equations with decimals.

Cluster 3: Geometry & Measurement (Questions 36 - 50)

No.	Question & Options	Ans	Source & Diagnostic Purpose
36	Sum of angles on a straight line is: A) 90° B) 180° C) 360° D) 270°	B	[P6/JSS1] Basic geometric theorems.
37	Calculate area of triangle: base=10cm, height=6cm. A) 60cm ² B) 30cm ² C) 16cm ² D) 32cm ²	B	[P6] Mensuration foundations.
38	An angle measuring 210° is described as: A) Obtuse B) Acute C) Reflex D) Right angle	C	[P6/JSS1] Classification of angles.
39	How many degrees are in 1/4 of a full turn? A) 45° B) 90° C) 180° D) 60°	B	[Cambridge P6] Rotational logic.
40	Perimeter of a square with side 7cm. A) 49cm B) 28cm C) 14cm D) 21cm	B	[P5] Perimeter vs Area confusion check.
41	Sum of interior angles of a triangle is: A) 90° B) 180° C) 360° D) 120°	B	[JSS1] Triangle properties.
42	Number of faces on a cube: A) 4 B) 6 C) 8 D) 12	B	[P5/6] 3D Shape properties.
43	Third angle in triangle with 40° and 60°: A) 100° B) 80° C) 20° D) 120°	B	[JSS1] Missing angle calculation.
44	Volume of cuboid: L=5cm, W=4cm, H=3cm. A) 12cm ³ B) 60cm ³ C) 20cm ³ D) 47cm ³	B	[P6/JSS1] 3D Measurement.
45	Lines that never meet even when extended are: A) Perpendicular B) Parallel C) Intersecting D) Vertical	B	[P5] Basic line definitions.
46	Circumference of a circle formula is: A) πr B) $2\pi r$ C) πr^2 D) $2r$	B	[JSS1] Measurement of circles.
47	Value of x if x and 50° are vertically opposite angles. A) 40° B) 50° C) 130° D) 180°	B	[JSS1] Intersecting lines properties.
48	Convert 2.5kg to grams. A) 250g B) 25g C) 2500g D) 25000g	C	[P5/6] Metric conversions.
49	A polygon with 6 sides is a: A) Pentagon B) Hexagon C) Heptagon D) Octagon	B	[P6] Shape naming convention.
50	Complimentary angle to 30° is: A) 60° B) 150° C) 70° D) 90°	A	[JSS1] Complementary vs Supplementary check.

Cluster 4: Data Handling (Questions 51 - 60)

No.	Question & Options	Ans	Source & Diagnostic Purpose
51	Mean of 4, 6, 8, 10: A) 7 B) 28 C) 8 D) 6	A	[P6] Calculating averages.
52	Mode of 2, 3, 3, 5, 7, 7, 7, 8: A) 3 B) 7 C) 8 D) 5	B	[P5/6] Understanding frequency.
53	Median of 1, 3, 5, 7, 9: A) 5 B) 4 C) 9 D) 3	A	[P6] Mid-point identification.
54	Probability of rolling a 6 on a fair die: A) 1 B) 1/2 C) 1/6 D) 6	C	[JSS1/Cambridge] Basic probability.

No.	Question & Options	Ans	Source & Diagnostic Purpose
55	Range of data: 12, 5, 20, 8, 15. A) 15 B) 12 C) 8 D) 20	A	[JSS1] Data spread calculation (20-5).
56	Total of angles in a pie chart representing whole data: A) 90° B) 180° C) 360° D) 100°	C	[JSS1] Data representation foundations.
57	If a coin is tossed, probability of Head is: A) 0 B) 0.5 C) 1 D) 2	B	[P6/JSS1] Likelihood assessment.
58	Difference between Highest and Lowest values in a set: A) Mean B) Median C) Mode D) Range	D	[P6] Statistical terminology.
59	Average of first 5 odd numbers: A) 5 B) 25 C) 3 D) 9	A	[JSS1 Reasoning] Multi-step calculation.
60	Diagnostic Insight: Difficulty with Cluster 2 & 3 usually indicates a student will struggle with JSS2 Higher Algebra and Trigonometry.	-	[Final Note] Pathway readiness indicator.